

Can we put a price on climate impacts? *Measuring and valuing damages*

ECON-4210

Economics of Climate Change

Class 10

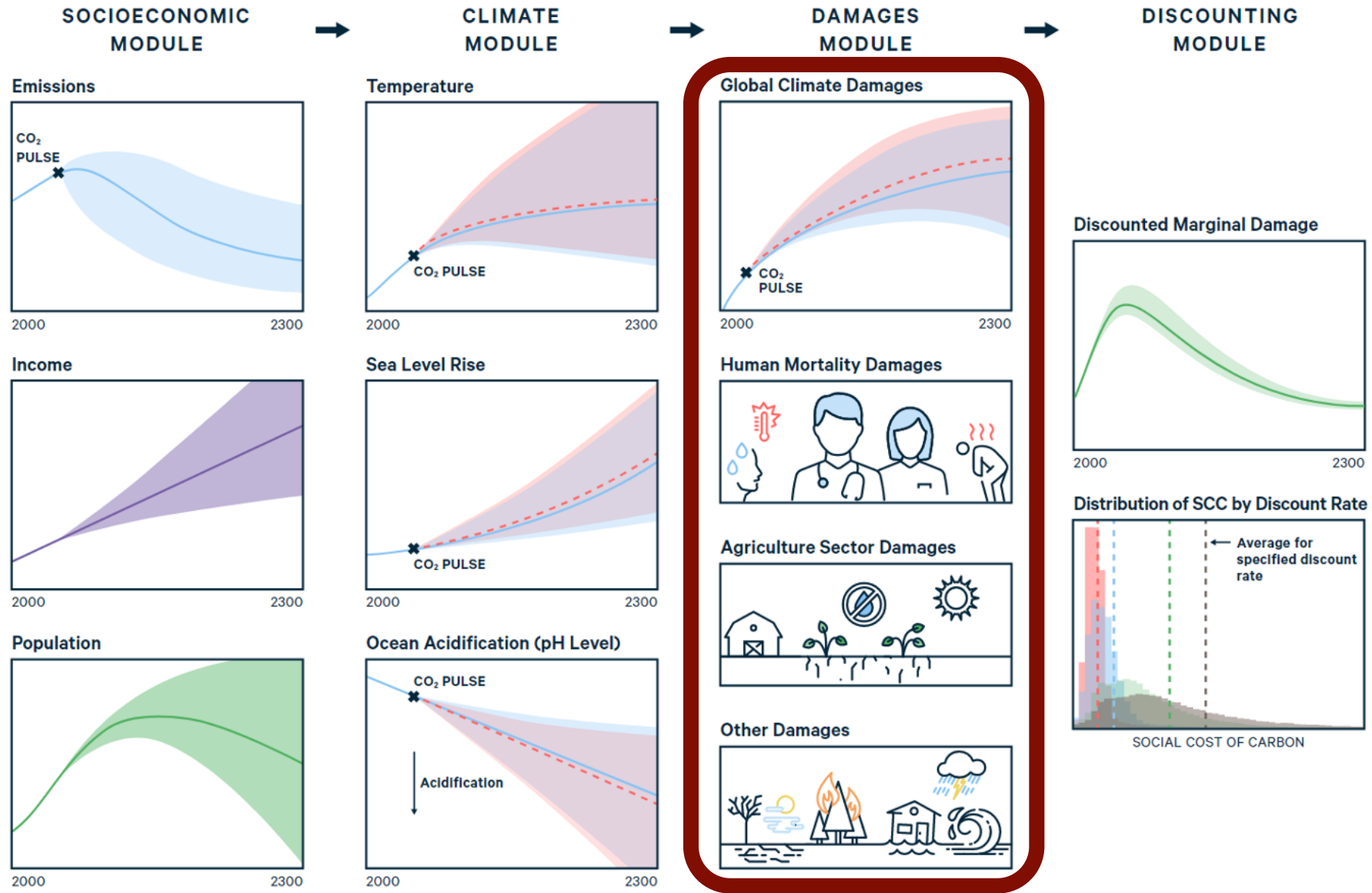
Prof. Casey Wichman

Today's class

1. Questions?
2. Attendance
3. Ghost Emissions
4. **Climate damage functions**
5. For next class...

What is a climate damage function?

Basic modeling framework used to estimate the social cost of carbon (SCC)



Climate damage functions in (old) SCC IAMs

DICE, FUND, & PAGE

- Generally, quadratic functions of global mean temperature
 - Includes non-temperature damages associated with increases in global mean temperatures.
- Very different structure across models.
- Damage estimates are based on older research
 - FUND: 2009 (mostly mid-1990s)
 - DICE: 1990s
 - PAGE: <2010
 - 100s of studies since then... none of which are represented in the “official” SCC.
- Newer studies (e.g., Rennert et al., 2022) have improved representation of damages...

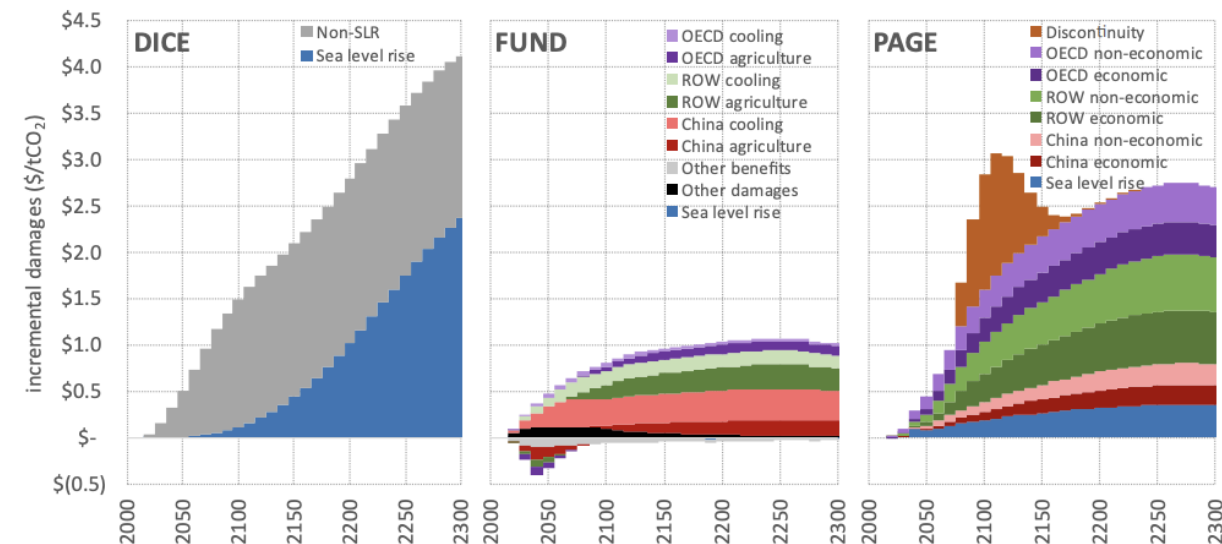
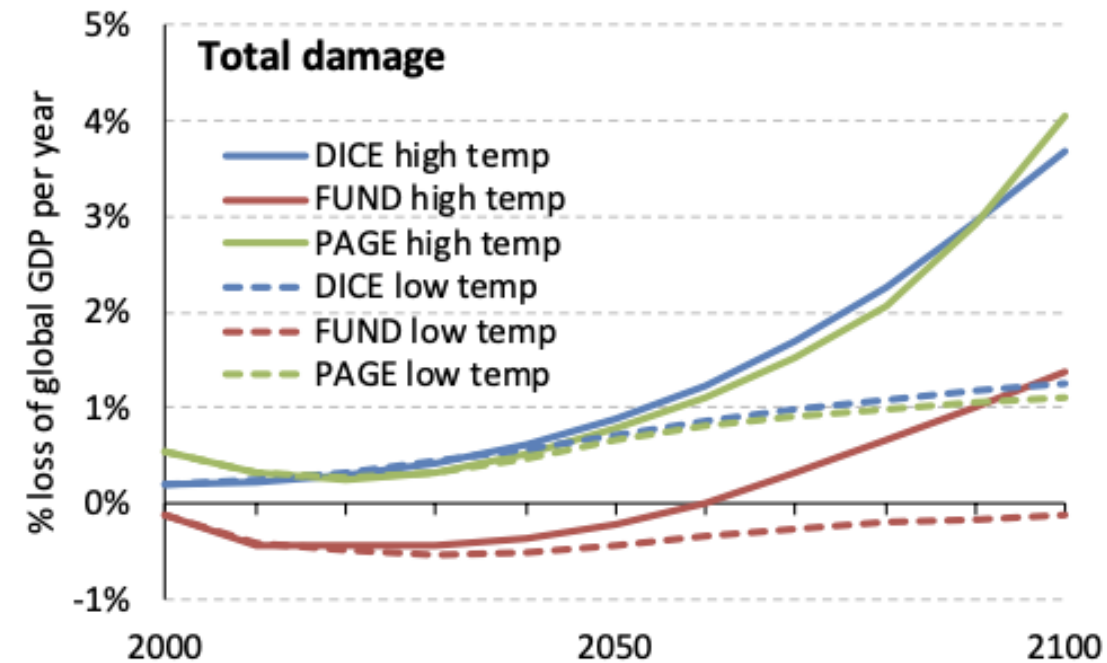


Figure 7. Key factors of annual incremental global climate damages to 2300 with the standardized high temperature and USG2 socioeconomic inputs.

Empirical damage functions

- The damage functions IAMs used to estimate the SCC rely on primary research from individual sectors (e.g., how will climate change affect agricultural crop yields.)

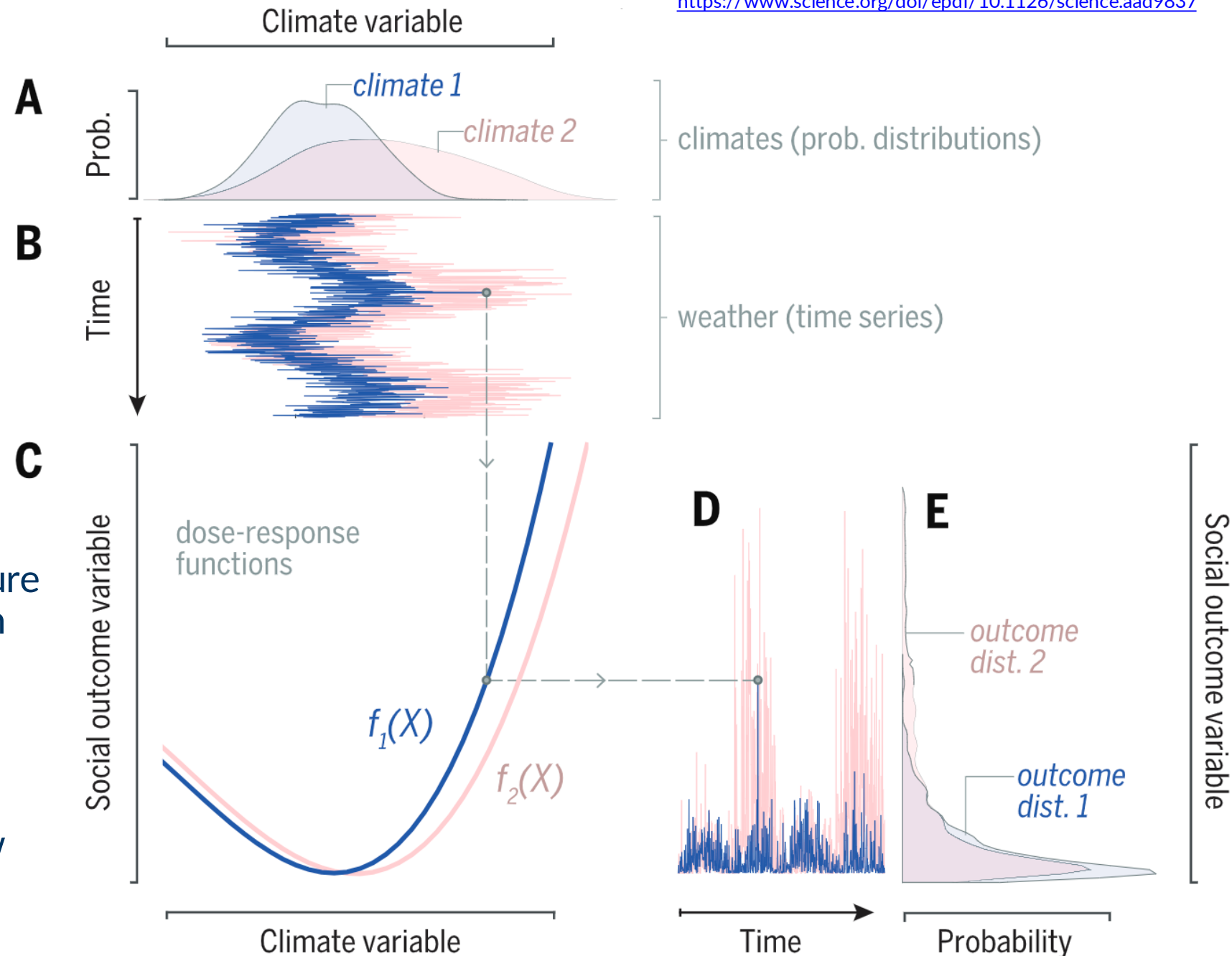
$$y = f(\Delta Climate)$$

- y is some economic outcome that we care about (e.g., agricultural output, heat-related mortality, electricity demand, etc.)
- Damage functions are very crude:
 - If studies suggest economic output is quadratic in climate change, we might use the following to represent climate damages:
$$y = 0.0T + 0.00236 * T^2$$
 - Where T represents global mean surface temperature change
 - This is the damage function in Bill Nordhaus' DICE model.

Climate vs. weather

- Remember: we can think of weather as a draw from the climate distribution
 - Climate 1 (blue) is current climate
 - Climate 2 (pink) is future climate.
- Researchers attempt to measure the “*dose-response function*” (in panel (c)) of some economic variable
- This can tell us how our outcome will change with new weather.

Source: Carleton & Hsiang (2016) —
<https://www.science.org/doi/epdf/10.1126/science.aad9837>



Estimating climate damages empirically

- For some outcomes (e.g., agricultural output or electricity demand), researchers use weather fluctuations to estimate climate change impacts.
 - E.g., we can use observed historical data to estimate the following relationship:

$$y^0 = f(W^0)$$

- This means we're trying to figure out how observed weather W^0 affects our outcome y^0 .
- The **weather-response function** $f(\cdot)$ could be linear, quadratic, very irregular, etc.
- Then, we use climate model output to understand what weather will look like in the future (W^1) and we plug this into the f function that we've estimated:

$$y^1 = f(W^1)$$

- So, *under certain conditions*, $y^1 - y^0$ provides the impact of climate change on our economic outcome y

Econometrics of climate change (i)

- How do researchers actually estimate $f(\cdot)$?
- Gather historical data on weather (W^0) (and outcome of interest (y)).
- Estimate relationship between y and W^0 econometrically.
- We can estimate this relationship using something like:

$$y = f(W^0) \Rightarrow y = \alpha + \beta W^0 + \varepsilon$$

- The coefficients (α, β) we estimate from this equation give us the functional form of f .
- Researchers choose the form of f , which can influence the ultimate outcome.

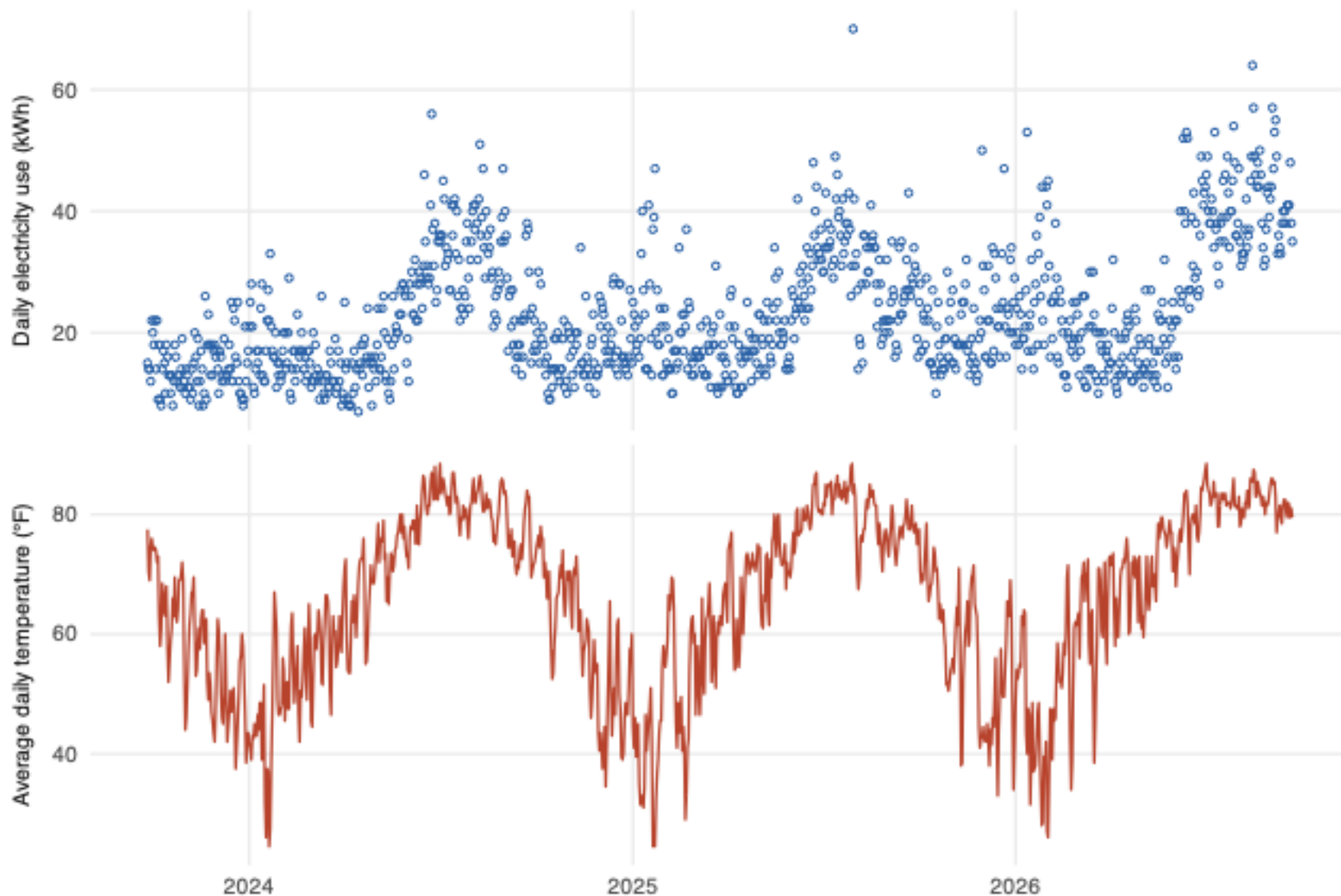
“Intuitive” explanation of econometrics...

$$y = \alpha + \beta W^0 + \varepsilon$$

- This is an OLS regression equation.
 - Regression attempts to explain how changes in an independent variable W^0 lead to changes in our dependent (or outcome) variable y .
- β is the coefficient that captures this relationship. In its simplest form this is just the correlation coefficient between W^0 and y .
 - Interpretation: A one-unit increase in W^0 leads to a β change in y .
- We’re just estimating a “best-fit” line of the form ($y = mx + b$) to the the data.
- We’ll work on interpreting regressions, but this will be helpful for understanding some upcoming readings.

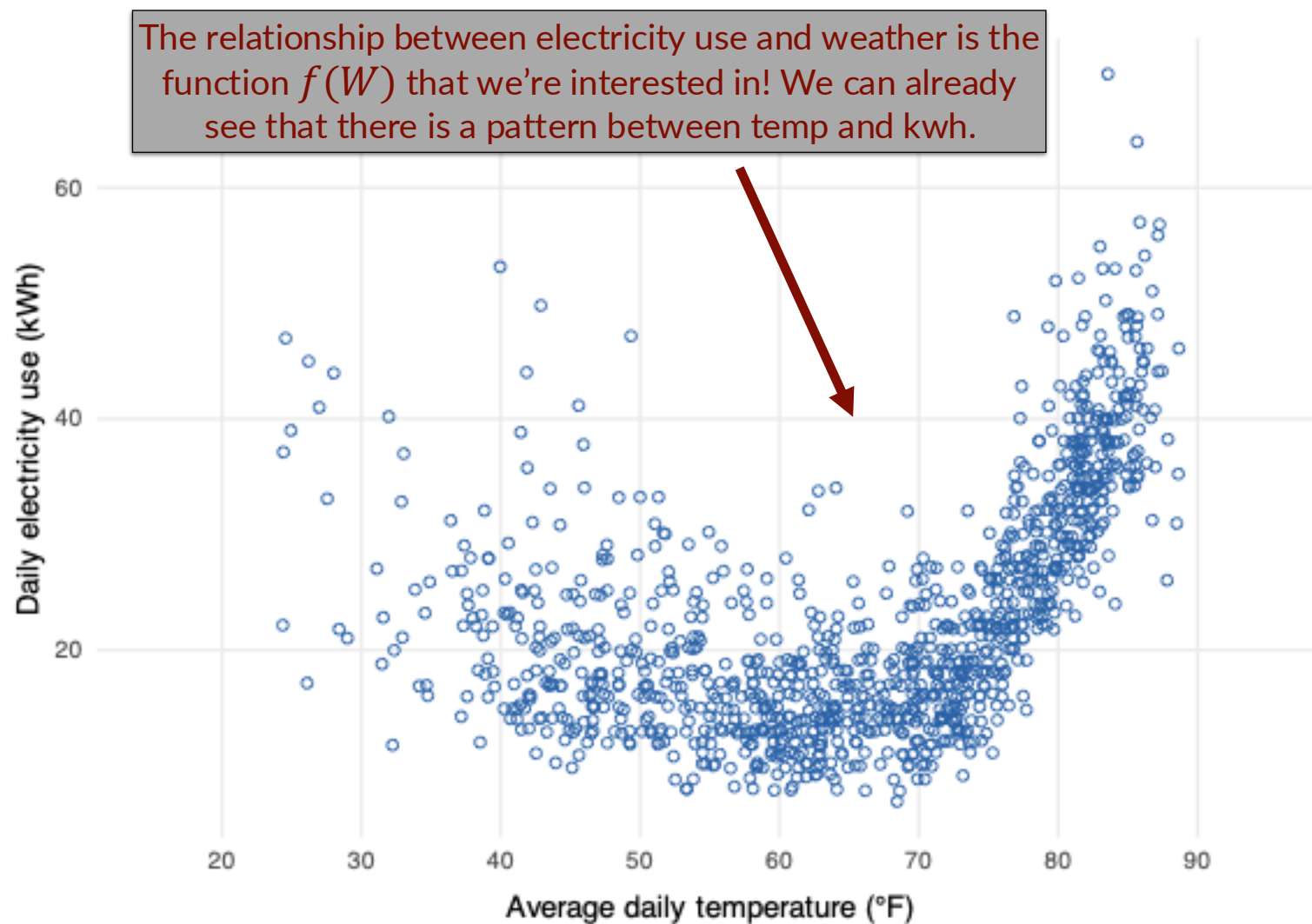
Estimating Prof. Wichman's weather-response function

- **Question:** how does my electricity use change with weather?
- **Data:**
 - Daily kWh usage since Sep 2023 (downloaded from Georgia Power)
 - Daily average outdoor temperatures.
- Obviously, electricity use is seasonal and responds to weather.
- I use more AC when it's hot out. I use more lights when it's dark out.



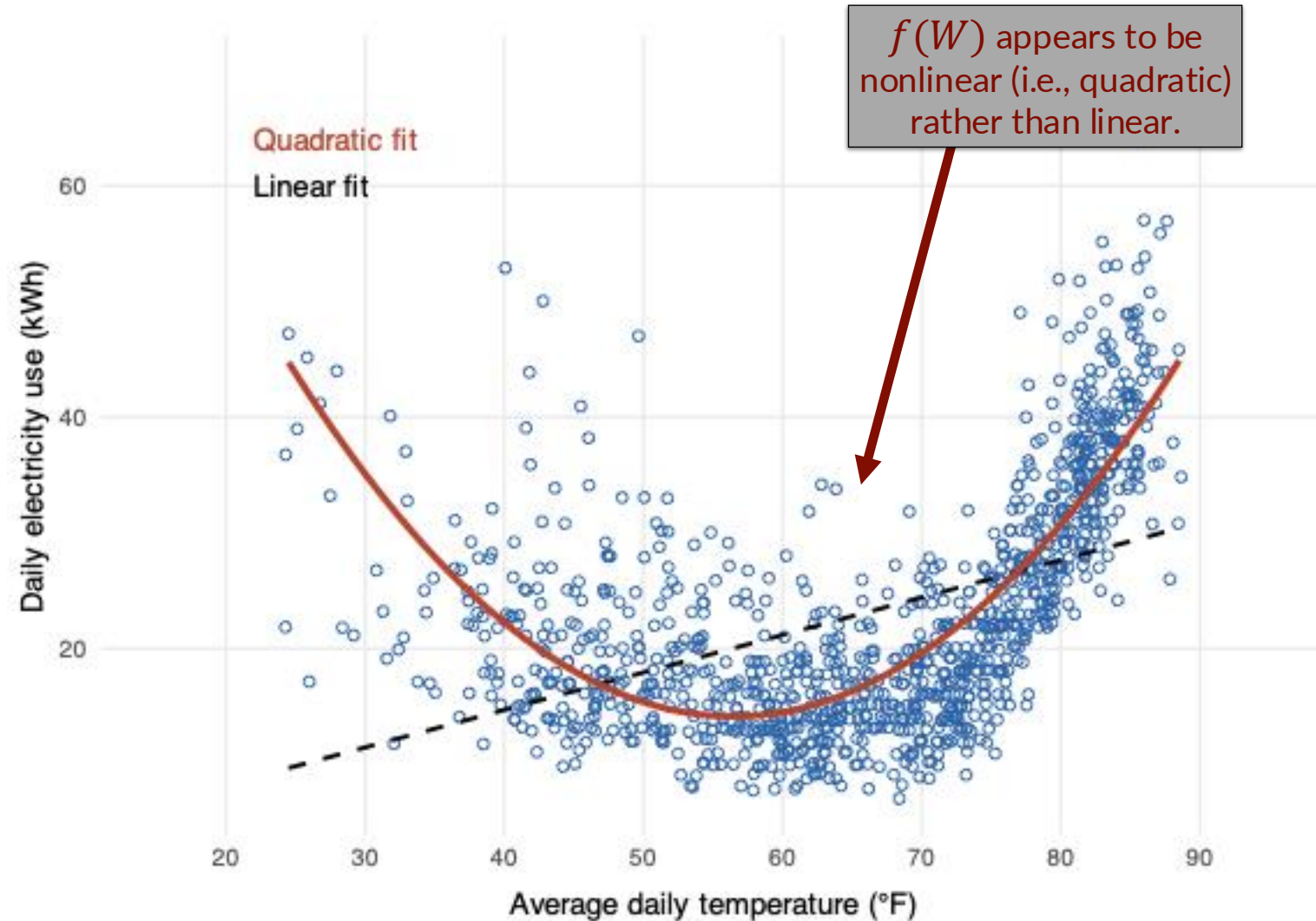
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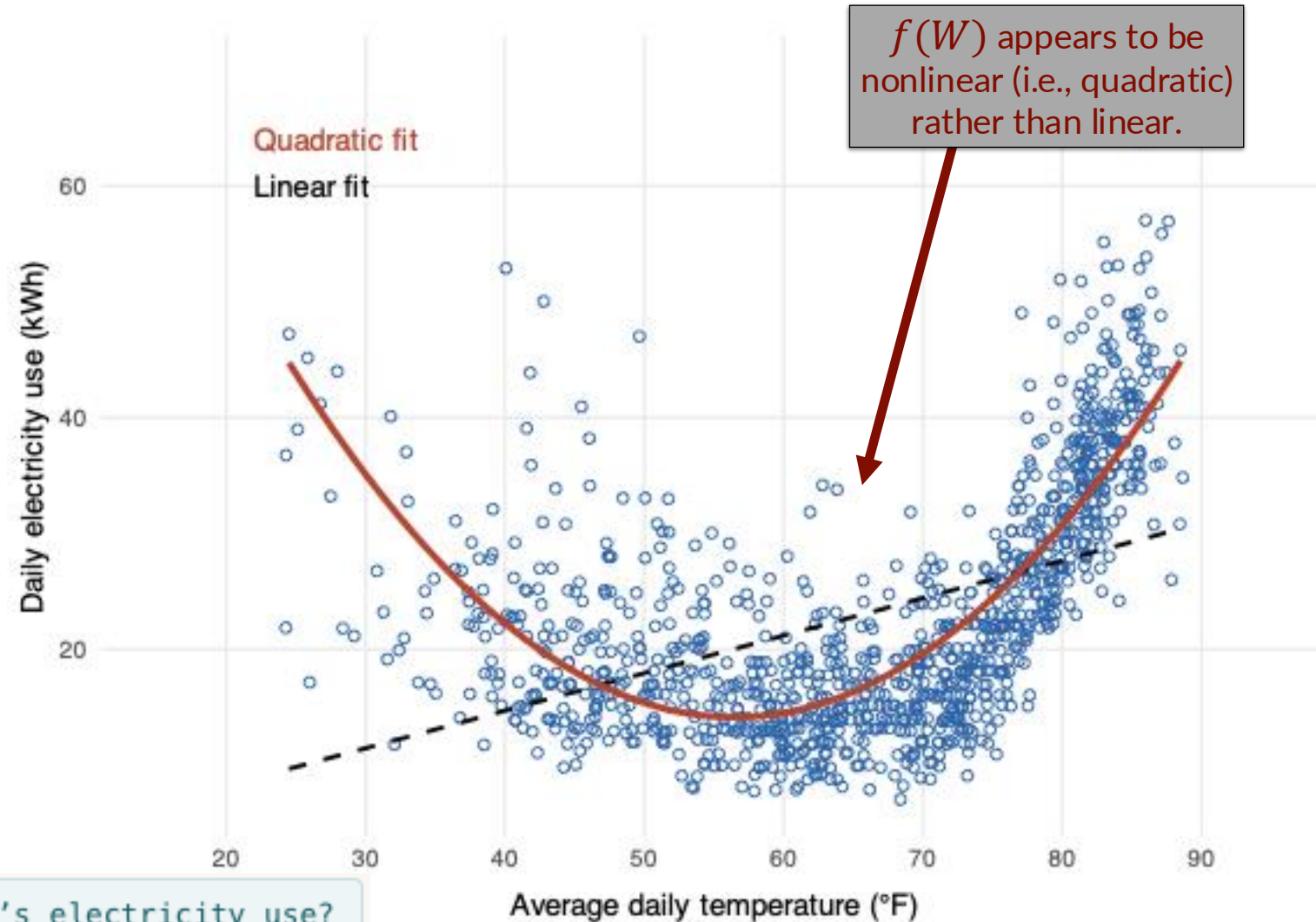
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Estimating Prof. Wichman's weather-response function

- How would climate change affect my electricity demand?



Econometrics of climate change (ii)

- That was an illustrative example, but not too far off what researchers are actually doing.
- Typically, we add a spatial cross-sectional dimension, too, so the equation becomes:

$$y_{it} = f(W_{it}^0)$$

- Where i might be counties (i =Fulton, Gwinnett, DeKalb...) or countries (i =US, Canada, Mexico...) and t represents time (e.g., years).
- We observe our outcome for each i and each t as well as weather in i and t .
- Researchers use both variation in weather over space (i.e., over i) and variation in weather over time (e.g., over t) to estimate f .

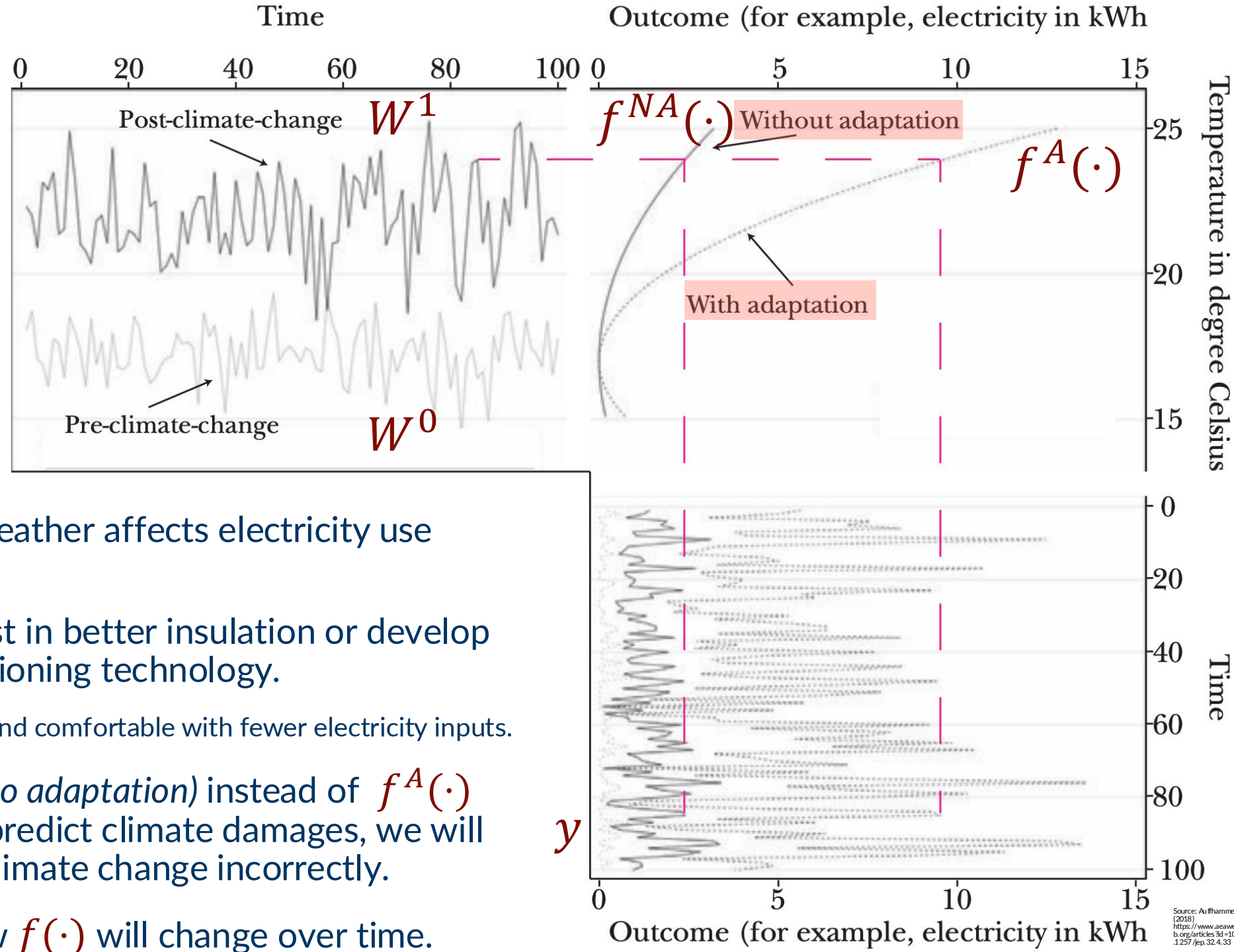
Econometrics of climate change (iii)

Thought experiment

- ▶ Imagine we've estimated $GDP_{it} = f(W_{it})$ for the US and Mexico between 1990 and 2020.
- ▶ We haven't observed the US economy in 2050, but we do observe the Mexican economy in 2010.
- ▶ If we take the hotter Mexican climate and give it to the US, could we predict how the US economy will respond in 2050?

Estimating climate damages with adaptation

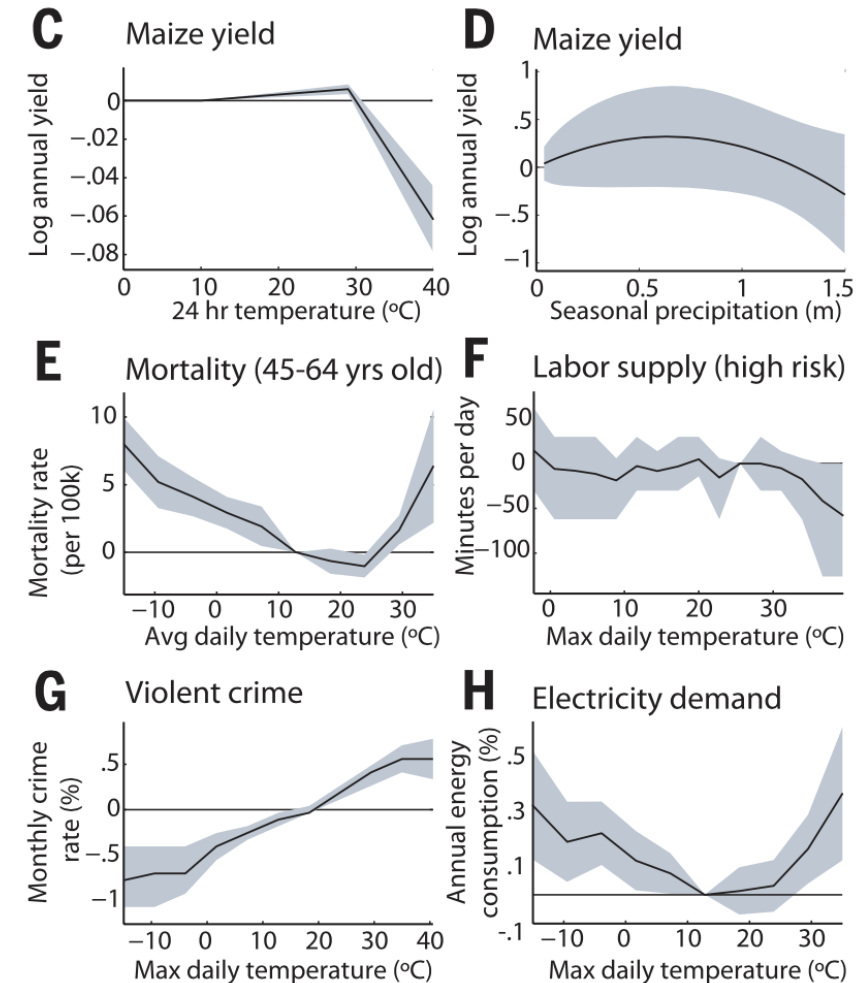
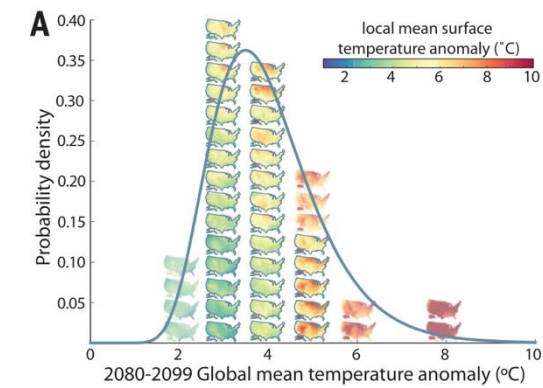
- We can estimate how weather affects electricity use today
- Over time, we may invest in better insulation or develop more efficient air-conditioning technology.
 - This allows us to stay cool and comfortable with fewer electricity inputs.
- If we use $f^{NA}(\cdot)$ (i.e., no adaptation) instead of $f^A(\cdot)$ (i.e., with adaptation) to predict climate damages, we will measure the impact of climate change incorrectly.
- It's difficult to know how $f(\cdot)$ will change over time.

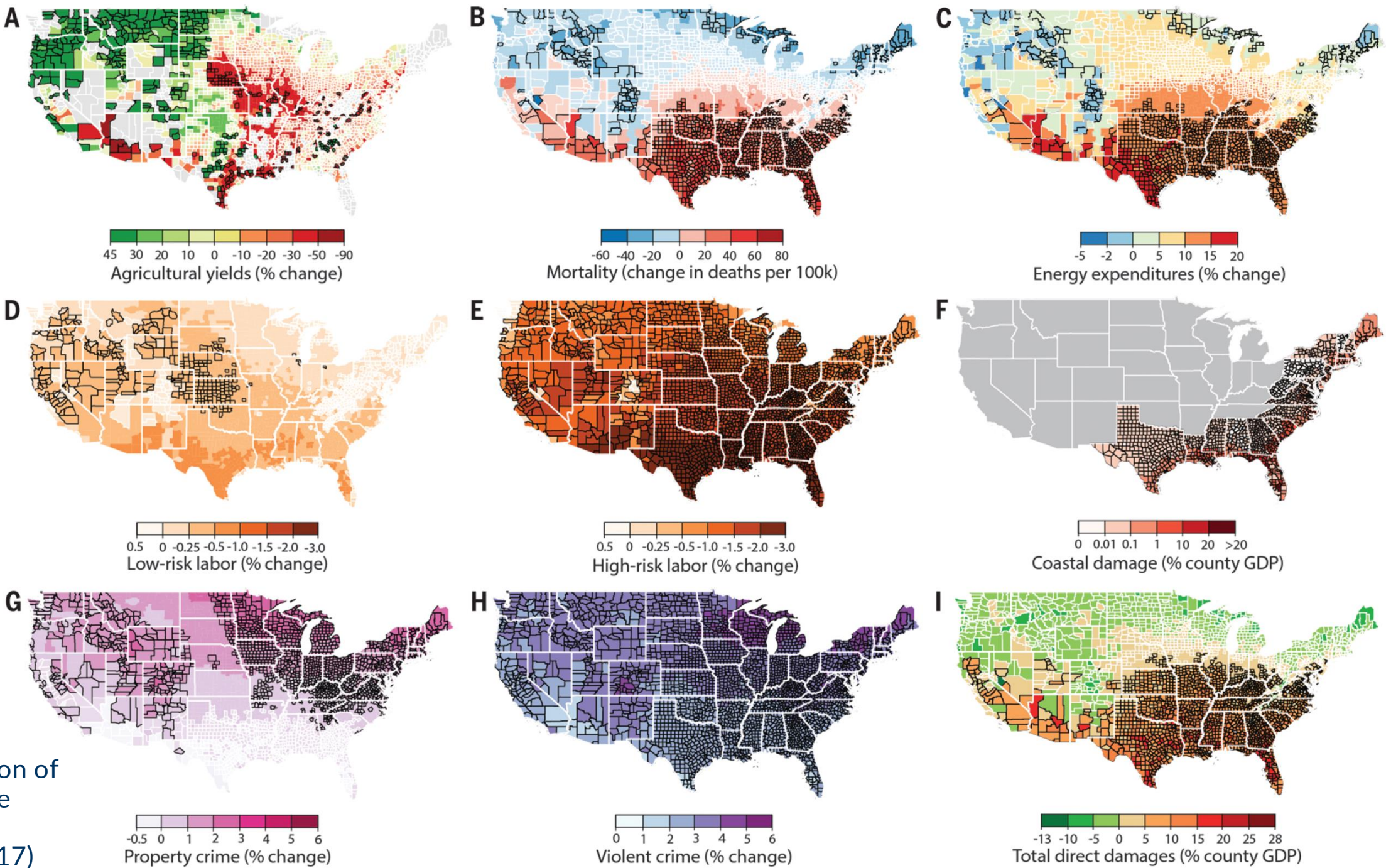


Estimates of US climate damages

Hsiang et al. (2017)

- Large-scale empirical damage estimation for several key outcomes (agriculture, mortality, labor supply, crime, electricity demand, coastal storms) in the US.
- Data for each county
- Projected out to 2080-2099 using (~3–5 deg C temperature change) using RCP8.5.
 - “Business-as-usual” scenario, which is fairly pessimistic.
- Attempt to measure a “bottom-up” damage function by adding up damages in each sector for each county in the US.



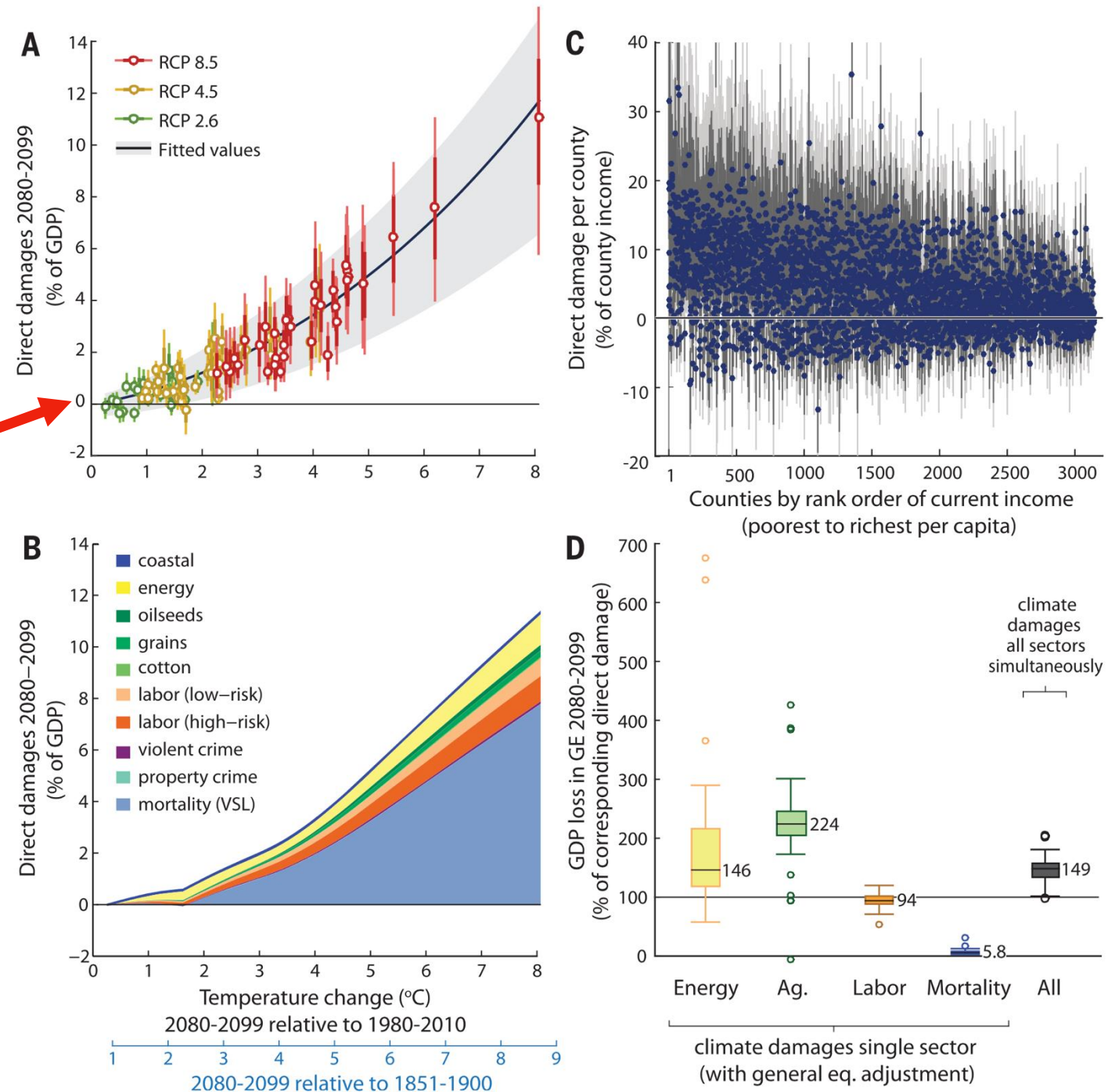


Spatial distribution of projected climate damages from Hsiang et al. (2017)

Estimates of US climate damages

Key takeaways from Hsiang et al. (2017):

- Damage = $0.283 \Delta T + 0.146 \Delta T^2$ 
- Damages $\approx$ 1.2% of GDP lost for every 1°C in global mean temperature.
- Poorest third of counties expected to see damages equal to 2–20% of county income by end of the century (under business-as-usual emissions).
- Key sectors included, but still doesn't include *everything*.



Some issues with econometric approaches to estimating climate damages

1. What if the weather-response function changes over time (e.g., because of adaptation, or technology change)?
2. Is a response to changes in “weather” really the same as a response to changes in “climate”?
3. We can only estimate weather-responses for sectors where we have lots of data (both across space and across time).
 - In the US and western Europe, we have pretty good data; less true in developing countries.
4. Most econometric approaches focus on hotter temperatures (and changes in precipitation), but climate change affects other dimensions of weather and extreme events (some of which are difficult to model)

See you on Wednesday!

caseyjwichman.com/econ4210/#week6